Exercice 1 : Zoo

Question 1

Entitée faible

Zoo(idZoo, nom)
Enclos(idEnclos, idZoo, nom, surface)

$$\text{Enclos}[\text{idZoo}] \subseteq \text{Zoo}[\text{idZoo}]$$

Cardinalité 1..1

Menu(idMenu, qteViande, qteLegume)
Famille(race, origine, ami)
Animal(idAnimal, nom, dateNaissance, race, origin, idMenu)

$$\begin{cases} \text{Animal}[\text{idMenu}] \subseteq \text{Menu}[\text{idMenu}] \\ \text{Animal}[(\text{race, origine})] \subseteq \text{Famille}[(\text{race, origine})] \end{cases}$$

Cardinalité 0..1

Animal(idAnimal, nom, dateNaissance, race, origin, idMenu)
Enclos(idEnclos, idZoo, nom, surface)

Occupe(idAnimal, idEnclos, idZoo)

$$\begin{cases} \text{Occupe}[\text{idAnimal}] \subseteq \text{Animal}[\text{idAnimal}] \\ \text{Occupe}[\text{idEnclos}] \subseteq \text{Enclos}[\text{idEnclos}] \end{cases}$$

Tables ajoutés

Ces tables ont étés ajoutés pour faire la relation parent enfant de l'Animal :
AnimalParent(idAnimalP, idAnimalE)
AnimalEnfant(idAnimalE, idAnimalP)

Question 2

```
CREATE TABLE Zoo (
    idZoo CHAR(5),
    nom VARCHAR(20),
    CONSTRAINT pk_Zoo PRIMARY KEY(idZoo));


CREATE TABLE Enclos (
    idEnclos CHAR(5), idZoo CHAR(5),
    nom VARCHAR(20), surface INTEGER,
    CONSTRAINT pk_Enclos PRIMARY KEY(idEnclos, idZoo),
    CONSTRAINT fk_Zoo FOREIGN KEY (idZoo) REFERENCES Zoo(idZoo));



CREATE TABLE Menu (
    idMenu CHAR(5),
    qteViande REAL,
    qteLegume REAL,
    CONSTRAINT pk_Menu PRIMARY KEY(idMenu));

CREATE TABLE Famille (
    race VARCHAR(20),
    origine VARCHAR(20),
    ami BOOLEAN,
    CONSTRAINT pk_Famille PRIMARY KEY(race, origine));


CREATE TABLE Animal (
    idAnimal CHAR(5),
    nom VARCHAR(20),
    dateNaissance DATE,
    race VARCHAR(20),
```

```
    origine VARCHAR(20),
    idMenu CHAR(5),
    CONSTRAINT pk_Animal PRIMARY KEY(idAnimal),
    CONSTRAINT fk_Animal_Famille FOREIGN KEY(race, origine) REFERENCES Famille(race, origine),
    CONSTRAINT fk_Animal_Menu FOREIGN KEY(idMenu) REFERENCES Menu(idMenu));

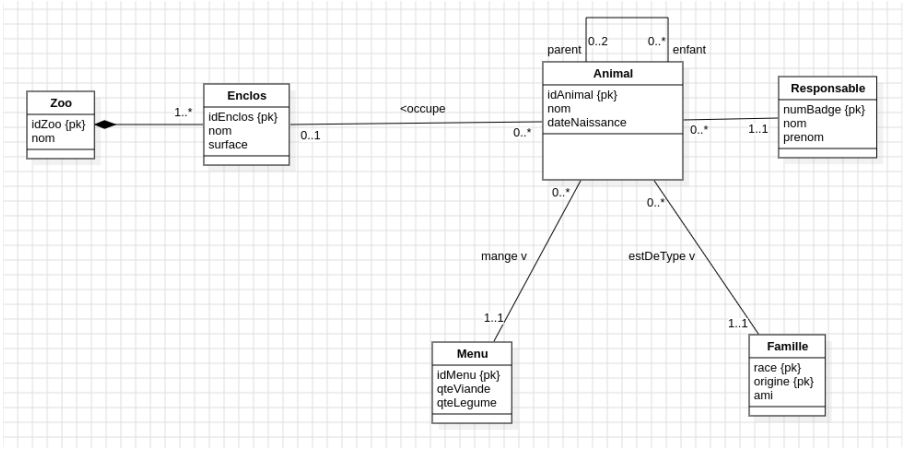

CREATE TABLE AnimalParent (
    idAnimalP CHAR(5),
    idAnimalE CHAR(5),
    CONSTRAINT pk_AnimalParent PRIMARY KEY(idAnimalP),
    CONSTRAINT fk_AnimalParent FOREIGN KEY(idAnimalP) REFERENCES Animal(idAnimal),
    CONSTRAINT fk_AnimalEnfant FOREIGN KEY(idAnimalE) REFERENCES Animal(idAnimal));


CREATE TABLE AnimalEnfant (
    idAnimalP CHAR(5),
    idAnimalE CHAR(5),
    CONSTRAINT pk_AnimalEnfant PRIMARY KEY(idAnimalE),
    CONSTRAINT fk_AnimalEnfant FOREIGN KEY(idAnimalE) REFERENCES Animal(idAnimal),
    CONSTRAINT fk_AnimalParent FOREIGN KEY(idAnimalP) REFERENCES Animal(idAnimal));




CREATE TABLE occupe (
    idAnimal CHAR(5),
    idEnclos CHAR(5),
    idZoo CHAR(5),
    CONSTRAINT pk_occupe PRIMARY KEY(idAnimal),
    CONSTRAINT fk_occupe_Animal FOREIGN KEY(idAnimal) REFERENCES animal(idAnimal),
    CONSTRAINT fk_occupe_Enclos FOREIGN KEY(idEnclos, idZoo) REFERENCES enclos(idEnclos, idZoo));
```

Question 3



Question 4

Responsable(<u>numBadge</u>, nom, prenom)
Animal(<u>idAnimal</u>, nom, dateNaissance, race, origin, idMenu, numBadge)

$$\text{Animal}[\text{numBadge}] \subseteq \text{Responsable}[\text{numBadge}]$$

Queston 5

```
CREATE TABLE Responsable (
    numBadge CHAR(5),
    nom VARCHAR(20),
    prenom VARCHAR(20),
    CONSTRAINT pk_responsable PRIMARY KEY(numBadge));

ALTER TABLE Animal ADD numBadge CHAR(5);

ALTER TABLE Animal ADD CONSTRAINT fk_Animal_Responsable FOREIGN KEY(numBadge) REFERENCES Responsable(numBadge);
```

Question 6

```
INSERT INTO Zoo VALUES ('Z1', 'Zoo de Valence');

INSERT INTO Enclos VALUES ('E1','Z1', 'Royaume des Lions', 2300);
INSERT INTO Enclos VALUES ('E2','Z1', 'Savane', 1200);

INSERT INTO Menu VALUES ('M1', 7.5, 0);
INSERT INTO Menu VALUES ('M2', 0, 40 );

INSERT INTO  Responsable VALUES('12345','DURAND', 'Olivier');
INSERT INTO  Responsable VALUES('56789','DUPOND', 'Nathalie');



INSERT INTO Famille VALUES('Lion', 'Afrique', false);
INSERT INTO Famille VALUES('Girafe', 'Afrique', true);

INSERT INTO Animal VALUES('L1', 'Roger', '2023-10-26', 'Lion', 'Afrique', 'M1', '12345');
INSERT INTO Animal VALUES('L2', 'Per', '2016-07-13', 'Lion', 'Afrique', 'M1', '56789');
INSERT INTO Animal VALUES('G2', 'Gunilla', '2021-03-05', 'Girafe', 'Afrique', 'M2', '56789');


INSERT INTO AnimalParent VALUES('L2', 'L1');
INSERT INTO AnimalEnfant VALUES('L2', 'L1');
```

Question 7

```
UPDATE Menu SET qteLegume = qteLegume + 2
WHERE idmenu='M2';
```

Question 8

```
DELETE FROM AnimalEnfant
WHERE idanimalP = 'L2' or idanimalE = 'L2';

DELETE FROM AnimalParent
WHERE idanimalP = 'L2' or idanimalE = 'L2';

DELETE FROM Animal
WHERE idanimal = 'L2';
```

Question 9

On change sélectionne une autre base de donnés :

```
\c postgres
```

On supprime la base de donnés :

```
DROP DATABASE "tp-zoo";
```

Exercice 2 : Les employés

Sélection, filtres, tri et jointures

Question 1

```
SELECT ename, hiredate FROM Emp
WHERE deptno=20;
```

```
 ename |  hiredate
-------+------------
 SMITH | 1980-12-17
 JONES | 1981-04-02
 SCOTT | 1982-12-09
```

```
 ADAMS | 1983-01-12
 FORD  | 1981-12-03
```

Question 2

```
SELECT ename, job, sal FROM Emp
WHERE sal>2800 and job='MANAGER';
```

```
 ename |   job   |   sal
-------+---------+---------
 JONES | MANAGER | 2975.00
 BLAKE | MANAGER | 2850.00
```

Question 3

```
SELECT ename, job, sal FROM Emp
WHERE deptno=30
ORDER BY sal ASC, ename DESC;
```

```
 ename  |   job    |   sal
--------+----------+---------
 JAMES  | CLERK    |  950.00
 WARD   | SALESMAN | 1250.00
 MARTIN | SALESMAN | 1250.00
 TURNER | SALESMAN | 1500.00
 ALLEN  | SALESMAN | 1600.00
 BLAKE  | MANAGER  | 2850.00
```

Question 4

```
SELECT ename, job, sal FROM Emp
WHERE sal>=1290 and ename LIKE 'M%';
```

```
 ename  |  job  |   sal
--------+-------+---------
 MILLER | CLERK | 1300.00
```

Question 5

```
SELECT ename FROM Emp
WHERE ename BETWEEN 'D' AND 'M'
ORDER BY ename;
```

```
 ename
-------
 FORD
 JAMES
 JONES
 KING
```

Question 6

```
SELECT ename, job, sal, d.deptno, dname, loc FROM Emp e, Dept d
WHERE e.deptno = d.deptno AND d.loc='DALLAS';
```

```
SELECT ename, job, sal, d.deptno, dname, loc FROM Emp e, Dept d
WHERE e.deptno = d.deptno AND d.loc='DALLAS';
 ename |   job   |   sal   | deptno |  dname   |  loc
-------+---------+---------+--------+----------+--------
 SMITH | CLERK   |  800.00 |     20 | RESEARCH | DALLAS
 JONES | MANAGER | 2975.00 |     20 | RESEARCH | DALLAS
 SCOTT | ANALYST | 3000.00 |     20 | RESEARCH | DALLAS
 ADAMS | CLERK   | 1100.00 |     20 | RESEARCH | DALLAS
 FORD  | ANALYST | 3000.00 |     20 | RESEARCH | DALLAS
```

Question 7

```
SELECT d.deptno, dname, loc FROM Emp e, Dept d
WHERE e.deptno = d.deptno AND e.job IN ('CLERK', 'SALESMAN','ANALYST');
```

```
 deptno |   dname    |   loc
--------+------------+----------
     10 | ACCOUNTING | NEW YORK
     20 | RESEARCH   | DALLAS
     20 | RESEARCH   | DALLAS
     20 | RESEARCH   | DALLAS
     20 | RESEARCH   | DALLAS
     30 | SALES      | CHICAGO
     30 | SALES      | CHICAGO
     30 | SALES      | CHICAGO
     30 | SALES      | CHICAGO
     30 | SALES      | CHICAGO
```

Question 8

```
SELECT ename, sal, d.deptno, loc FROM Emp e, Dept d
WHERE e.deptno = d.deptno AND loc IN ('DALLAS', 'CHICAGO') AND sal>=1000;
```

```
 ename  |   sal   | deptno |   loc
--------+---------+--------+---------
 ALLEN  | 1600.00 |     30 | CHICAGO
 WARD   | 1250.00 |     30 | CHICAGO
 JONES  | 2975.00 |     20 | DALLAS
 MARTIN | 1250.00 |     30 | CHICAGO
 BLAKE  | 2850.00 |     30 | CHICAGO
 SCOTT  | 3000.00 |     20 | DALLAS
 TURNER | 1500.00 |     30 | CHICAGO
 ADAMS  | 1100.00 |     20 | DALLAS
 FORD   | 3000.00 |     20 | DALLAS
```

Question 9

```
SELECT empno, ename, hiredate FROM Emp
WHERE hiredate<'1981-11-30'
ORDER BY ename;
```

```
 empno | ename  |  hiredate
-------+--------+------------
  7499 | ALLEN  | 1981-02-20
  7698 | BLAKE  | 1981-05-01
  7782 | CLARK  | 1981-06-09
  7566 | JONES  | 1981-04-02
  7839 | KING   | 1981-11-17
  7654 | MARTIN | 1981-09-28
  7369 | SMITH  | 1980-12-17
  7844 | TURNER | 1981-09-08
  7521 | WARD   | 1981-02-22
```

Sous-requêtes et hiérarchie

Question 10

```
SELECT ename, sal FROM Emp as employee
WHERE sal>(
SELECT sal FROM Emp as manager
WHERE employee.mgr=manager.empno)
;
```

```
 ename |   sal
-------+---------
```

```
 SCOTT | 3000.00
 FORD  | 3000.00
```

Question 11

```
SELECT ename FROM Emp as employee
WHERE hiredate<(
SELECT hiredate FROM Emp as manager
WHERE employee.mgr=manager.empno)
;
```

```
 ename
-------
 SMITH
 ALLEN
 WARD
 JONES
 BLAKE
 CLARK
```

Question 12

```
SELECT d.dname FROM Dept AS d
WHERE NOT EXISTS (
    SELECT * FROM Emp AS e
    WHERE e.deptno = d.deptno AND e.job = 'ANALYST'
);
```

```
   dname
------------
 ACCOUNTING
 SALES
 OPERATIONS
```

Question 13

```
SELECT ename FROM Emp
WHERE empno IN (
    SELECT empno FROM Emp
    WHERE mgr IN (
        SELECT empno FROM Emp
        WHERE job='CLERK'
        )
        INTERSECT
    SELECT empno FROM Emp
    WHERE mgr IN (
        SELECT empno FROM Emp
        WHERE job='SALESMAN'
        )
)
;
```

```
 ename
-------
```

Question 14

```
SELECT ename, sal FROM Emp
WHERE sal>(
    SELECT MAX(sal) FROM Emp
    WHERE ename='JONES' or ename='MARTIN'
);
```

```
 ename |   sal
-------+---------
```

```
 SCOTT | 3000.00
 KING  | 5000.00
 FORD  | 3000.00
```

Question 15

```
    SELECT dname FROM Dept as d
    WHERE NOT EXISTS (
        SELECT empno FROM Emp as e
        WHERE e.deptno = d.deptno
    )
    ;
```

```
   dname
------------
 OPERATIONS
```

Question 16

```
SELECT ename FROM Emp as employee
WHERE EXISTS (
    SELECT empno FROM Emp as e, Dept as d
    WHERE employee.sal>e.sal
    AND e.deptno = d.deptno
    AND d.deptno = '30'
);
```

```
 ename
--------
 ALLEN
 WARD
 JONES
 MARTIN
 BLAKE
 CLARK
 SCOTT
 KING
 TURNER
 ADAMS
 FORD
 MILLER
```

Question 17

```
SELECT ename FROM Emp
WHERE empno IN (
    SELECT DISTINCT mgr FROM Emp
);
```

```
 ename
-------
 JONES
 BLAKE
 CLARK
 SCOTT
 KING
 FORD
```

Question 18

```
SELECT empno, ename, job, sal, comm FROM Emp
WHERE empno IN (
    SELECT mgr FROM Emp
    WHERE comm IS NOT NULL
);
```

```
 empno | ename |   job   |   sal   | comm
-------+-------+---------+---------+------
  7698 | BLAKE | MANAGER | 2850.00 |
```

Question 19

```sql
SELECT
e.ename AS nom,
e.sal AS salaire,
s.grade AS echelle_salaire,
manager.ename AS superieur,
d.deptno AS num_dept,
d.loc AS ville_dept
FROM Emp AS e
LEFT OUTER JOIN Salgrade AS s
ON e.sal BETWEEN s.losal AND s.hisal
LEFT OUTER JOIN Emp AS manager
ON e.mgr = manager.empno
LEFT OUTER JOIN Dept AS d
ON e.deptno = d.deptno
ORDER BY e.sal, e.ename
;
```

```
  nom   | salaire | echelle_salaire | superieur | num_dept | ville_dept
--------+---------+-----------------+-----------+----------+------------
 SMITH  |  800.00 |               1 | FORD      |       20 | DALLAS
 JAMES  |  950.00 |               1 | BLAKE     |       30 | CHICAGO
 ADAMS  | 1100.00 |               1 | SCOTT     |       20 | DALLAS
 MARTIN | 1250.00 |               2 | BLAKE     |       30 | CHICAGO
 WARD   | 1250.00 |               2 | BLAKE     |       30 | CHICAGO
 MILLER | 1300.00 |               2 | CLARK     |       10 | NEW YORK
 TURNER | 1500.00 |               3 | BLAKE     |       30 | CHICAGO
 ALLEN  | 1600.00 |               3 | BLAKE     |       30 | CHICAGO
 CLARK  | 2450.00 |               4 | KING      |       10 | NEW YORK
 BLAKE  | 2850.00 |               4 | KING      |       30 | CHICAGO
 JONES  | 2975.00 |               4 | KING      |       20 | DALLAS
 FORD   | 3000.00 |               4 | JONES     |       20 | DALLAS
 SCOTT  | 3000.00 |               4 | JONES     |       20 | DALLAS
 KING   | 5000.00 |               5 |           |       10 | NEW YORK
```

Agrégats et regroupement

Question 20

```sql
SELECT d.deptno, d.dname, d.loc
FROM Dept AS d
NATURAL RIGHT OUTER JOIN Emp AS e
GROUP BY d.deptno, d.dname, d.loc
HAVING COUNT(e.empno) > 3 AND d.loc != 'CHICAGO'
;
```

```
 deptno |  dname   |  loc
--------+----------+--------
     20 | RESEARCH | DALLAS
```

Question 21

```sql
SELECT d.dname, COUNT(e.empno) AS nbre_max_employes
FROM Dept AS d
NATURAL RIGHT OUTER JOIN Emp AS e
GROUP BY d.dname
HAVING COUNT(e.empno)=(
    SELECT MAX(nbre_emp_per_dept)
    FROM (
        SELECT COUNT(*) AS nbre_emp_per_dept
        FROM Emp
```

```
        GROUP BY deptno
        )
    )
;
```

```
 dname | nbre_max_employes
-------+------------------
 SALES |                 6
```

Question 22

Facile :

```
SELECT manager.ename,
COUNT(manager.ename) AS Nb_subordonnes
FROM Emp AS manager
LEFT OUTER JOIN Emp AS subordonnes
ON manager.empno = subordonnes.mgr
WHERE subordonnes.mgr IS NOT NULL
GROUP BY manager.ename
ORDER BY Nb_subordonnes
;
```

```
 ename | nb_subordonnes
-------+----------------
 FORD  |              1
 CLARK |              1
 SCOTT |              1
 JONES |              2
 KING  |              3
 BLAKE |              5
```

Difficile :

```
SELECT manager.ename,
COUNT(subordonnes.mgr) AS Nb_subordonnes
FROM Emp AS manager
LEFT OUTER JOIN Emp AS subordonnes
ON manager.empno = subordonnes.mgr
GROUP BY manager.ename
ORDER BY Nb_subordonnes
;
```

```
 ename  | nb_subordonnes
--------+----------------
 ALLEN  |              0
 MARTIN |              0
 JAMES  |              0
 ADAMS  |              0
 TURNER |              0
 MILLER |              0
 SMITH  |              0
 WARD   |              0
 CLARK  |              1
 FORD   |              1
 SCOTT  |              1
 JONES  |              2
 KING   |              3
 BLAKE  |              5
```

Question 23

```
SELECT
d.dname,
e.job,
SUM(e.sal) AS Total_salaire,
COUNT(e.empno) AS Nb_employes,
```

```
AVG(e.sal) AS Salaire_moyen

FROM Dept as d

LEFT OUTER JOIN Emp as e
ON d.deptno = e.deptno

GROUP BY d.dname, e.job
HAVING COUNT(e.job)>=2
;
```

```
  dname   |   job    | total_salaire | nb_employes |     salaire_moyen
----------+----------+---------------+-------------+-----------------------
 SALES    | SALESMAN |       5600.00 |           4 | 1400.0000000000000000
 RESEARCH | ANALYST  |       6000.00 |           2 | 3000.0000000000000000
 RESEARCH | CLERK    |       1900.00 |           2 |  950.0000000000000000
```

Question 24

```
SELECT
d.dname,
e.job,
SUM(e.sal) AS Total_salaire,
COUNT(e.empno) AS Nb_employes,
AVG(e.sal) AS Salaire_moyen

FROM Dept as d

LEFT OUTER JOIN Emp as e
ON d.deptno = e.deptno

GROUP BY d.dname, e.job

ORDER BY Nb_employes DESC, d.dname, e.job
;
```

```
   dname    |    job    | total_salaire | nb_employes |     salaire_moyen
------------+-----------+---------------+-------------+-----------------------
 SALES      | SALESMAN  |       5600.00 |           4 | 1400.0000000000000000
 RESEARCH   | ANALYST   |       6000.00 |           2 | 3000.0000000000000000
 RESEARCH   | CLERK     |       1900.00 |           2 |  950.0000000000000000
 ACCOUNTING | CLERK     |       1300.00 |           1 | 1300.0000000000000000
 ACCOUNTING | MANAGER   |       2450.00 |           1 | 2450.0000000000000000
 ACCOUNTING | PRESIDENT |       5000.00 |           1 | 5000.0000000000000000
 RESEARCH   | MANAGER   |       2975.00 |           1 | 2975.0000000000000000
 SALES      | CLERK     |        950.00 |           1 |  950.0000000000000000
 SALES      | MANAGER   |       2850.00 |           1 | 2850.0000000000000000
 OPERATIONS |           |               |           0 |
```

Question 25

```
SELECT EXTRACT(YEAR FROM e.hiredate) AS Annee, COUNT(e.empno) AS Nb_Employes
FROM Emp AS e
GROUP BY Annee
ORDER BY Annee
;
```

```
 annee | nb_employes
-------+-------------
  1980 |           1
  1981 |          10
  1982 |           2
  1983 |           1
```

Question 26

Facile :

```
SELECT
d.dname,
AVG(e.sal) AS Salaire_moyen
FROM Dept AS d
LEFT OUTER JOIN Emp e
ON d.deptno = e.deptno
WHERE e.sal IS NOT NULL
GROUP BY d.dname
;
```

```
   dname    |     salaire_moyen
------------+-----------------------
 ACCOUNTING | 2916.6666666666666667
 SALES      | 1566.6666666666666667
 RESEARCH   | 2175.0000000000000000
```

Difficile :

```
SELECT
d.dname,
CONCAT(ROUND(AVG(e.sal), 2), ' dollars') AS Salaire_moyen
FROM Dept AS d
LEFT OUTER JOIN Emp e
ON d.deptno = e.deptno
WHERE e.sal IS NOT NULL
GROUP BY d.dname
;
```

```
   dname    |  salaire_moyen
------------+-----------------
 ACCOUNTING | 2916.67 dollars
 SALES      | 1566.67 dollars
 RESEARCH   | 2175.00 dollars
```

Question 27

Facile :

```
SELECT
d.dname,
SUM(e.sal) AS SALAIRE,
SUM(e.comm) AS COMMISSION
FROM Dept AS d
LEFT OUTER JOIN Emp e
ON d.deptno = e.deptno
GROUP BY d.dname
;
```

```
   dname    | salaire  | commission
------------+----------+------------
 ACCOUNTING |  8750.00 |
 SALES      |  9400.00 |    2200.00
 OPERATIONS |          |
 RESEARCH   | 10875.00 |
```

Difficile :

```
SELECT
d.dname,
CONCAT(COALESCE(SUM(e.sal), '0.00'), ' dollars') AS SALAIRE,
CONCAT(COALESCE(SUM(e.comm), '0.00'), ' dollars') AS COMMISSION
FROM Dept AS d
LEFT OUTER JOIN Emp e
```

```
ON d.deptno = e.deptno
GROUP BY d.dname
;
```

```
   dname    |     salaire      |   commission
------------+------------------+-----------------
 ACCOUNTING | 8750.00 dollars  | 0.00 dollars
 SALES      | 9400.00 dollars  | 2200.00 dollars
 OPERATIONS | 0.00 dollars     | 0.00 dollars
 RESEARCH   | 10875.00 dollars | 0.00 dollars
```

Question 28

```
SELECT
    TABLE_MIN.dname AS Moins_cher,
    TABLE_MIN.salaire_min AS mini,
    TABLE_MAX.dname AS Plus_cher,
    TABLE_MAX.salaire_max AS maxi
FROM (
     SELECT d.dname, SUM(e.sal) AS salaire_min
     FROM Dept d
     JOIN Emp e ON d.deptno = e.deptno
     GROUP BY d.deptno, d.dname
     HAVING SUM(e.sal) = (
         SELECT MIN(total_sal)
         FROM (
             SELECT SUM(sal) AS total_sal
             FROM Emp
             GROUP BY deptno
         )
     )) TABLE_MIN,
    (
     SELECT d.dname, SUM(e.sal) AS salaire_max
     FROM Dept d
     JOIN Emp e ON d.deptno = e.deptno
     GROUP BY d.deptno, d.dname
     HAVING SUM(e.sal) = (
         SELECT MAX(total_sal)
         FROM (
             SELECT SUM(sal) AS total_sal
             FROM Emp
             GROUP BY deptno
         )
     )) TABLE_MAX;
```

```
 moins_cher |  mini   | plus_cher |   maxi
------------+---------+-----------+----------
 ACCOUNTING | 8750.00 | RESEARCH  | 10875.00
```